Deva Anand

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Education

University of Massachusetts Amherst — **September 2025 – May 2027 (Expected)**
Master of Science in Computer Science — Amherst, MA
Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies — **August 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering — Chennai, India

Skills

Languages	Python, C++, JavaScript, TypeScript, Java, SQL, Bash
Agentic AI & Evals	LLM agents, agentic workflows, tool use, verifiable / auditable evaluation, anti-Goodharting guardrails, prompt design, OpenAI / Groq APIs
ML & Deep Learning	PyTorch, JAX, NumPy, autodiff (from scratch), generative models, normalizing flows, diffusion, HuggingFace Transformers
Distributed & Systems	Apache Spark / PySpark, distributed-data processing, high-throughput / low-latency tuning, durable workflows, async services, Docker
Backend & Cloud	FastAPI, Node.js, REST APIs, multi-tenant systems, AWS, GCP, Azure, Git, GitHub Actions, CI/CD, Linux

Experience

Wysa — **July 2023 – July 2025**
Backend Engineer — Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd. — **November 2022 – May 2023**
Backend Developer Intern — Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input; iterated on prompt design and response-quality heuristics for a real customer-facing experience while building 20+ Node.js + Fastify REST APIs over PostgreSQL.

Projects

AgenticSearch - Provenance-First Entity Discovery — github.com/Deva-1903/ciir_agentic_search
Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables using typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

AutoEval - Verifiable, Self-Improving Evaluation — github.com/Deva-1903/AutoEval
Python, evaluation methodology, Git automation

- Built a self-improving evaluation agent that proposes test cases and labels, scores them against multiple systems, and accepts only updates that improve **discriminative power** on a holdout-heavy objective.
- Added **anti-Goodharting guardrails** (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates to Git with report cards, a **verifiable, reproducible** eval workflow.

cf-ai-research-scout - Edge-Native AI Research Assistant — github.com/Deva-1903/cf_ai_research_scout
TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an edge-native AI research assistant: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step tasks survive process restarts and tool failures, with conversation state persisted in D1.

Spark ETL - Distributed Processing at Scale — github.com/Deva-1903/Spark-ETL-Optimization
Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Built a distributed pipeline over a **1.4B-row** dataset, cutting runtime **25%** and tail-latency ratio from **2.14x to 1.74x** via broadcast joins, adaptive execution, and partition tuning, distributed-systems work at real scale.

Publication & Extracurriculars

- **Publication:** *"Alzheimer's Disease Classification using Transfer Learning,"* **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.